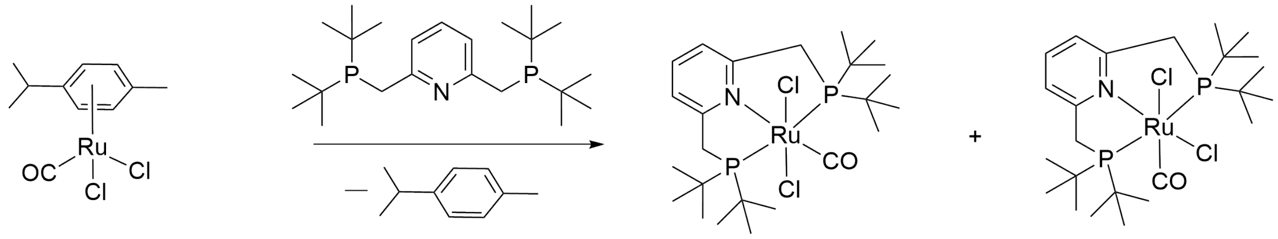
AE-232: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

Date: 2024-03-06
Tags: [RuCl2(CO)(p-cymene)] Dichloro PNP [RuCl2(CO)(PNP)] P(tBu)N(py)P(tBu) NMR AE 1H 31P
Status: Done
Created by: Alexander Eith

Reaction scheme/sample structure



THF, reflux, o.n.

$[RuCl_2(CO)(P^{tBu}N^{py}P^{tBu})]$

$[RuCl_2(CO)(p\text{-cymene})]$

Molecular Weight: 334.20 g/mol

2,6-bis((di-*tert*-butylphosphaneyl)methyl)pyridine
Chemical Formula: $C_{23}H_{43}NP_2$
Molecular Weight: 395,55152

Chemical Formula: $C_{24}H_{43}Cl_2NOP_2Ru$
Molecular Weight: 595,53152

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
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Related experiments	Experiment - AEI-060: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)], Experiment - AEI-072: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)], Experiment - AE-108: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)], Experiment - AE-157: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)], Experiment - AE-230: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

Reagents

Name	Amount [mmol]	Equivalents	Purity	$Mass_{theo,pure}$ [mg]	$Mass_{theo}$ [mg]	$Mass_{exp}$ [mg]	Molar mass [g/mol]	Volume [ml]
$[RuCl_2(CO)(p\text{-cymene})]$ (Experiment - AE-221: Synthesis of [RuCl2(CO)(p-cymene)]	0.593	1.00	0.58	198.19	347.72	341.92	334.20	/
PNP, 2,6-Bis(N,N-(tert-butyl)phosphinomethyl)pyridine (Experiment - AE-219: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine	0.919	1.55	0.90	363.6	404	404	395.55	/
THF (Experiment - KRA-080: Degassing of THF	/	/	/	/	/	/	/	33

n-heptane (Experiment - AE-120: Degassing of n-Heptane	/	/	/	/	/	/	/	

Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an argon filled glovebox. (see Protocol Protocol - Schlenk Technique)

Date	Time	Step	Observations
06.03		A 50 mL Schlenk flask was prepared	
		The air cooler was dried by setting under vacuum	approx. 2 h
	13:00	PNP(tBu) was added to the flask in a glovebox.	white solid
	13:30	[RuCl2(CO)(p-cymene)] was weighted in air into the 50 mL Flask and the flask was set under argon.	red solid [Ru]
	15:00	To the flask THF (33 mL) was added.	
		The reaction mixture was stirred at rt and the flask was conected to the cooler in argon counterflow.	A red suspension is obtained
07.03	15:30 - 9:00	The reaction mixture was heated to reflux for approx. 17:30 h at approx. 250 rpm.	
	- 13:25	The reaction mixture was cooled down to r.t.	
	13:25 - 13:35	The volume of the remaining reaction mixture was reduced under reduced pressure to approx. 1/2	after reducing the volume.jpg
		To the solution n-Heptane (33 mL) was added under stirring.	
		The obtained suspension was stored at -20 °C	
08.03	14:50	After warming to room temperature: the obtained mixture was filtered according to Filtration with frit technique and the residue was washed with n-Heptane (2 * 3 mL and 2 * 2 mL)	after filtration.jpg
	15:10 - 15:40	The solid (AE-232-1) was dried under reduced pressure.	prod.jpg

11.03		A NMR-sample was prepared from AE-232-1 according to [Protocol] Preparation of NMR Sample	NMR sample.jpg
25.03	10:35	The remaining soild after Experiment - AE-239: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h was dissolved in THF (2 * 8 mL) and transffered in to a 50 mL flask	in THF.jpg
	10:45	n-heptane (16 mL) was added	after addition of n-hept.jpg
	11:10	The mixture was filtered according to Filtration with frit technique and the residue was washed with n-Heptane (3 * 1 mL)	
	12:25 - 12:45	The obtained solid was dried under reduced pressure	prod.jpg
	13:35	An NMR sample was prepared according to Protocol - Preparation of NMR Sample in DCM-d2	-2 NMR sample.jpg

Analysis

Date	Time	Sample name	Analysis method	Used device	Solvent	Raw Data	Processed Data	Comparative data	Interpretation
12.03	/	AE-232-1	1H, 31P(quant)	IAAC 400 MHz I	DCM-d2	AE-232-1-1H.zip , AE-232-1-31P.zip	AE-232-1_10.nmrium	Experiment - AE-230: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)] -2 Experiment - AEI-060: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)] -5-2 (not in the file)	Mainly desired product was obtained. But significant impurity eith approx. 20 % observed at 42 ppm. The impurity was already observed as major impurtiy in prior synthesises. (Like AEI-060) but in lower amounts. But it seems, that the impurtiy is not present after the silver oxide route (at least for syns. done with AEI-060, but also spectra were only measured in water.
27.03	/	AE-232-2	1H, 31P(quant)	IAAC 400 MHz I	DCM-d2	AE-232-2.zip	AE-232-2_20.nmrium	Experiment - AE-230: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)] -2	Desired product. No significant amounts of side products observed (maybe one species is observed, but to small for integration)

Product characterization

	mass [mg]	purity [%]	$mass_{pure}$ [mg]	amount [mmol]	Yield [%]	
AE-232-1	397.82	80	318.26	0.53	90	sand yellow solid, prod.jpg

AE-232-2	138.24	>98	138.24	0.23	not applicable, since material was removed from AE-232-1 for a reaction	yellow-greenish solid, prod.jpg

Linked experiments

- JSC-KS-04: Degassing of n-Heptane

- AEI-047: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine

- AEI-057: Synthesis of [RuCl2(CO)(p-cymene)]

- AEI-060: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AEI-072: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AE-100: Synthesis of [RuCl2(CO)(p-cymene)]

- AE-101: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AE-102: Degassing of THF

- AE-108: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AE-120: Degassing of n-Heptane

- AE-157: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- KRA-080: Degassing of THF

- AE-219: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine

- AE-221: Synthesis of [RuCl2(CO)(p-cymene)]

- AE-230: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AE-239: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

Linked resources

Protocol - Preparation of NMR Sample

Protocol - Filtration with frit technique

Protocol - Schlenk Technique

Attached files

AE-232-2_20.nmrium
sha256: c3fefb790263ce0eb522106f06b2171258dedd99df7ef0fc400b93616116eb67

AE-232-2.zip
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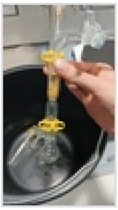

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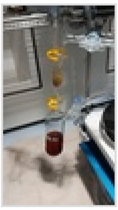

after-reducing-the-volume.jpg
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Link: https://elab.water-splitting.org/experiments.php?mode=view&id=867